

Ø<Traditional IT (Virtualization se pehle)

1. Physical Servers Only

Har ek application ke liye alag physical server lagana padta tha.*

Example: Agar tumhe Mail Server, Web Server, Database Server chahiye, to teen alag physical machines lagengi.*

2. Problems

Costly !' Har application ke liye naya hardware.*

Under-utilization !' Server ki CPU/RAM fully use nahi hoti thi (10–20% hi chalti thi).*

Maintenance heavy !' Cooling, power, space, security sab zyada chahiye.*

Scalability issue !' Naya application chahiye to naya server kharidna padta tha.*

Ø=Virtualization ka Arrival

Virtualization ne ek physical server ko multiple virtual machines (VMs) me tod diya.*

Ek server me hi alag-alag Virtual Machines bna kar tum multiple applications chala sakte ho.*

Har VM apna Operating System + Application run karta hai jaise wo alag server ho.*

Ø=Benefits:

Hardware utilization zyada (60–80% tak)*

Kam cost !' ek server pe multiple workloads*

Fast provisioning !' naya VM minutes me ban jata hai*

*Easy backup & recovery**

Tools / Hypervisors: VMware, Hyper-V, KVM

Ø<Cloud IT (Virtualization ke baad agla step)

Cloud ne virtualization ko aur ek level upar le gaya:

1. **On-Prem (Traditional Data Center) !'** Sab kuch khud manage karna.

2. **IaaS !'** Cloud tumhe VMs ready deta hai.

3. **PaaS** !' Cloud tumhe pura platform deta hai (sirf code likhna hai).
4. **SaaS** !' Cloud tumhe pura software ready deta hai (sirf use karna hai).

*Traditional IT** !' Har bache ke liye alag-alag laptop kharidna. (Costly, zyada jagah lagti)

*Virtualization** !' Ek hi bada laptop me multiple users ke liye alag-alag virtual desktops banana.

*Cloud** !' Laptop bhi tumhe kharidne ki zaroorat nahi, bas browser kholo aur ready software use karo